

Molecular Dynamics Simulation of Proton Transport in Polymer Electrolyte Membrane

Takuya Mabuchi^{1,*} and Takashi Tokumasu²

¹Graduate School of Engineering, Tohoku University, Sendai, Miyagi 980-8577, Japan

²Institute of Fluid Science, Tohoku University, Sendai, Miyagi 980-8577, Japan

We have performed a detailed analysis of proton solvation and transport properties in hydrated Nafion using molecular dynamics simulation. The revised empirical valence bond (EVB) method was developed in order to treat the excess proton transport through the Grotthuss mechanism. The new EVB model predicts a significantly enhanced transport in comparison with previous hopping models as well as the classical hydronium diffusion, which largely improves the agreement with the available experimental data. Our results suggest that a proton hopping mechanism has a small effect on the proton dissociation from the first solvation shell of sulfonate groups, namely that protons are not enhanced to separate from the sulfonate groups by the hopping mechanisms. From diffusion comparison between the Grotthuss and vehicular mechanism, the Grotthuss mechanism dominates the proton diffusion at the studied hydration levels including a hydration level of 3. It was also found that the vehicular mechanism dominates the electroosmotic transport of water molecules at the studied hydration levels.

Keywords: Empirical Valence Bond, Molecular Dynamics, Polymer Electrolyte, Transport Phenomena.

1. INTRODUCTION

Polymer electrolyte fuel cells (PEFCs) are highly expected as a next-generation power supply system due to the purity of its exhaust gas, its high power density and high efficiency. The heart of PEFCs is polymer electrolyte membrane (PEM) that separated the reactant gases and conducts protons. In the nanoscopic structure of the membrane, ion clusters are formed by water molecules gathered vicinity of sulfonate groups which are hydrophilic parts of Nafion membrane, and proton transport are largely attributed to the nanoscopic structure of polymer membranes and water aggregations. Thus, many macroscopic simulations that are based on continuum theory have found it difficult to discuss the relation between the structural and dynamical properties of protons within the membrane in the previous experimental research.¹ It is critical to understand the proton transport mechanisms through polymer membrane and clarify an important link between the membrane nanostructure and the proton transport properties.

In the bulk aqueous solution, it is observed that proton mobility is roughly 5 times higher than other cations such

as K^+ which have an ionic radius similar to the hydronium ions. This property of proton transport is attributed to a combination of proton hopping, known as the Grotthuss mechanism,² and vehicular mechanism where protons diffuse in solution as a hydronium ion, the smallest stable hydrated form of the excess proton in water. While proton transport through the Grotthuss mechanism is believed to involve the inter-conversion between two different solvation structures, the Eigen³ ($[H_2O \cdot (H_2O)_3]^+$) cation and Zundel⁴ ($[H \cdot (H_2O)]^+$) cation,^{5,6} the details of the hopping mechanisms is still debated. The theoretical basis for the proton hopping model has been investigated using the two-state empirical valence bond (EVB) method^{7,8} and the multistate empirical valence bond (MS-EVB) method^{9,10} which are the development of the empirical valence bond (EVB) approach originally formulated by Warshel and Weiss.¹¹

In Nafion membranes, it is considered that proton hopping transport still takes place although the water dynamic property in the bulk aqueous solution is inhibited by a lack of bulklike water structure at low water contents. Proton transport properties including hopping mechanism in hydrated polymer membrane have been studied using the

*Author to whom correspondence should be addressed.

quantum hopping (Q-HOP) MD method^{12,13} and the MS-EVB method.¹⁴ In their studies, there is a restriction of hoping of a single proton among water molecules while the remaining protons residing on hydronium ions were not allowed to hop during the simulation. This limitation can have an influence on proton dynamics especially at low hydration levels. The self-consistent multistate empirical valence bond (SCI-MS-EVB) methodology^{15,16} has also been used to capture the phenomena of multiproton systems, giving very deep insight and valuable results of proton mobility in hydrated Nafion. However, because the MS-EVB method requires 8–10 EVB states per proton in combination with several water molecules as large as 4, it prohibits the efficient application of the model to investigate the influence of proton hops under confined environments at very low hydration levels. In this study, a revised EVB model has been developed based on the previous study of the two-state EVB model reported by Walbran et al.⁸ We have investigated the impact of the proton hopping mechanism on the structure and dynamics properties of protons in hydrated Nafion membrane at different hydration levels as low as 3.

The paper is organized as follows: Section 2 provides details of the simulation methods. Section 3 discusses the simulation results for the analysis of proton solvation and transport properties. Section 4 summarizes our findings.

2. SIMULATION METHODS

The EVB approach, which is based on the previous study of EVB model reported by Walbran et al.⁸ has been used in this work to capture the underlying physics of multiproton systems. In the newly developed EVB model, we have considered a simplified cluster model of the Zundel cation where a hydronium ion exchanges a proton with a neighboring water molecule. Within a single cluster, we have considered a function of the proton transfer coordinate, defined as

$$Q \equiv |R_{O^*H}| - |R_{OH}| \quad (1)$$

where O^* and O are the oxygen of the hydronium ion and water molecule, respectively. Also, a total potential energy of the $H_5O_2^+$ complex V_{ad} uses the following expression:

$$V_{ad} = \frac{(V_i + V_f) - \sqrt{(V_i - V_f)^2 + 4\Lambda(Q, R)^2}}{2} \quad (2)$$

where i and f stand for the “initial” and “final” states of the cluster. In the initial state, V_i is the energy of the cluster calculated under the assumption that the molecule labeled “ H_3O^+ ” is treated as the hydronium model while the molecule labeled “ H_2O ” is considered as the water model as shown in Figure 1(a). In the final state, “ H_3O^+ ” is separated from the central proton and treated as the water model whereas the “ H_2O ” is combined with its proton and considered as the hydronium model as shown in Figure 1(b).

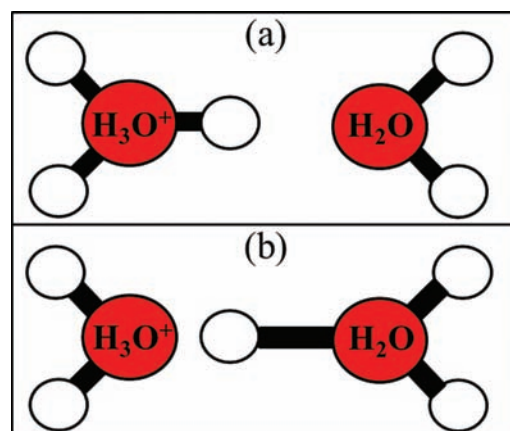


Figure 1. (a) Initial state and (b) final state of the $H_5O_2^+$ complex at the same position of the central proton. The molecule that oxygen atom (red circle) has three bonding (solid black line) with hydrogen atoms (white circle) interacts as in the hydronium model.

$\Lambda(Q, R)$ is a function of proton transport variable Q defined in Eq. (1) and R defined as the distance between O^* and O and is taken to be equal to 0 for $Q < -Q_0$ and $Q > Q_0$. For Q in the range $-Q_0 < Q < Q_0$ the function Λ is taken to be a following function

$$\Lambda(Q, R) = \left\{ 1 - \left(\frac{Q}{|Q_0|} \right)^2 \right\} \cdot F(Q, R) \quad (3)$$

and a function $F(Q, R)$ is a polynomial equation as follows

$$F(Q, R) = \alpha(R) \cdot \left(\frac{Q}{|Q_0|} \right)^4 + \beta(R) \cdot \left(\frac{Q}{|Q_0|} \right)^2 + \gamma(R) \quad (4)$$

$$\alpha(R) = \alpha_1 \cdot R^3 + \alpha_2 \cdot R^2 + \alpha_3 \cdot R + \alpha_4 \quad (5)$$

$$\beta(R) = \beta_1 \cdot R^3 + \beta_2 \cdot R^2 + \beta_3 \cdot R + \beta_4 \quad (6)$$

$$\gamma(R) = \gamma_1 \cdot R^3 + \gamma_2 \cdot R^2 + \gamma_3 \cdot R + \gamma_4 \quad (7)$$

V_{ad} is modified by parameterizing the function Λ , to fit it to the curves of the potential energy surface (PES) calculated by *ab initio* calculation.⁹ The final set of parameters for the EVB model is listed in Table I. Figure 2 shows a comparison between the *ab initio* data⁹ and the results obtained with the EVB model for O^*-O separations corresponding to 2.2, 2.4, 2.6, and 2.8 Å, respectively. In all 4 distance cases, the results of EVB model are in good agreement with the corresponding *ab initio* data.

Table I. Parameters for the EVB Model.

Q_0	0.5						
α_1	-1877.4	α_2	14732.6	α_3	-38310.0	α_4	33030.5
β_1	-213.8	β_2	1679.6	β_3	-4316.7	β_4	3625.8
γ_1	-15.1	γ_2	194.8	γ_3	-655.3	γ_4	698.9

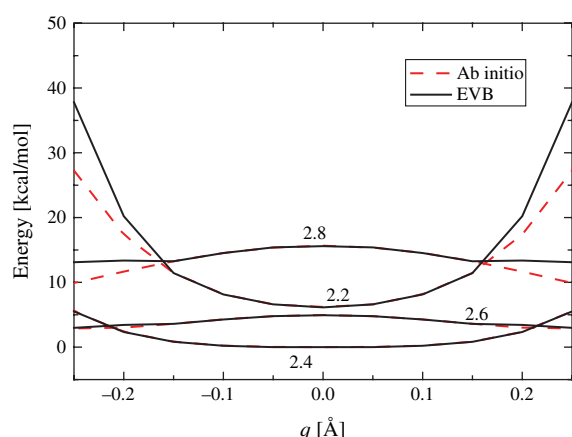


Figure 2. Potential energy surface (PES) curves for the proton shuttling in the Zundel cation at O*-O distances of 2.2, 2.4, 2.6, and 2.8 Å. Note that the proton shuttle coordinate, q , is defined as half the value of Q .

Nafion membrane, which has the chemical structure shown in Figure 3 with EW = 110 ($n = 7$ and $m = 1$), has been employed as a PEM in the simulation. We generated a Nafion chain by linking the polar monomeric unit (P) to the end of the nonpolar monomeric unit N_7 , repeating the procedure 13 times to obtain $(N_7P)_{13}$. The potential model for Nafion was based on the DREIDING force field as referred to in previous studies by other authors.^{17–20} The aSPC/Fw model⁹ for both hydronium ions and water molecules are employed. The initial configurations were generated by randomly placing 25 Nafion chains in a box, $x \times y \times z = 200 \times 200 \times 200 \text{ Å}^3$, with periodic boundary condition in all directions. To ensure charge neutrality, we added a total of 325 hydronium ions and considered water contents of $\lambda = 3, 6$, and 12, where the parameter λ indicates the ratio of the number of water molecules to that of SO_3^- ($N_{\text{H}_2\text{O}, \text{H}_3\text{O}^+}/N_{\text{SO}_3^-}$). After all molecules were added, the annealing procedure using the vehicular model (the classical hydronium model¹⁷) was carried out to eliminate the initial configuration of molecules, establishing the equilibrium state. Our annealing procedure was also based on a combination of a number of annealing methods proposed previously.^{17, 21–24} The simulation cell was compressed gradually for 50 ps to the experimental density²⁵ for each hydration level, and *NPT* MD simulation for 100 ps at 800 K and 100 MPa using the L-J potential parameters, ϵ , reduced to 1/100 magnitude among Nafion atoms.²¹ Then, the *NPT* ensemble for 50 ps was performed with decreasing temperature and pressure

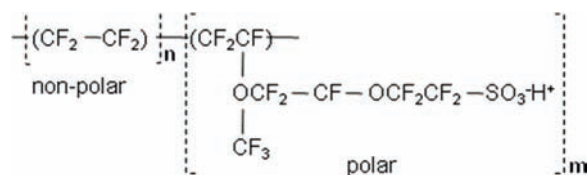


Figure 3. Chemical structure of Nafion membrane.

gradually from 800 to 300 K and 100 MPa to 0 MPa, respectively. In addition, we annealed them in four steps as follows:

1. *NPT* MD simulation for 100 ps at 300 K;
2. *NVT* MD simulation for 50 ps with the temperature increased gradually from 300 to 800 K;
3. *NVT* MD simulation for 50 ps at 800 K;
4. *NVT* MD simulation for 50 ps with the temperature decreased gradually from 800 to 300 K.

These four steps were repeated four times and the final density was attained after *NPT* MD simulation for 5 ns was utilized to relax the system to the desired pressure of 1 atm and temperature of 300 K. The *NVT* ensemble at 300 K was then performed to collect a total of 1 ns simulation data for the vehicular model. The final structures for the vehicular model were then adopted as the initial structures in the EVB simulations. For the EVB system, an extra *NVT* ensemble for 500 ps was employed to relax the structures under the EVB potential. The final structures for the EVB model were then used for the *NVT* ensemble of 1 ns to collect simulation data for the EVB model. For each system, the trajectory of molecules was collected every 0.5 ps. The temperature was held by the Nosé-Hoover thermostat,²⁶ and the pressure was controlled by the Andersen method.²⁷ The smooth particle mesh Ewald method²⁸ was used to calculate electrostatic interactions.

3. RESULTS AND DISCUSSION

Information on the proton solvation structure around sulfonate groups was obtained by the quantitative analysis of the excess proton distribution using the radial distribution function (RDF) shown as

$$g_{A-B}(r) = \left(\frac{n_B}{4\pi r^2 dr} \right) / \left(\frac{N_B}{V} \right) \quad (8)$$

where $4\pi r^2 dr$ is the volume of a dr thick spherical shell at radius r from particle A, n_B is the number of B particles in the shell, V is the volume of the system, and N_B is the number of B particles in the whole system. Figure 4 shows the RDFs, $g_{S-O_h}(r)$, between the sulfur atom of the sulfonate group and the oxygen atom of the hydronium ion ($S-O_h$) for the EVB model (solid line) and vehicular model (dashed line) at each water content. For both models $g_{S-O_h}(r)$ shows a strong narrow peak at 4 Å with the first minimum at ~ 4.3 Å. Thus, 4.3 Å is used to define the first solvation shell of sulfonate groups. Figure 5 shows the coordination number of hydronium ions in the first solvation shell n_{ch1} , for the EVB model (black square) and vehicular model (red circle) at each hydration level. The results show that n_{ch1} for both models decreases with increasing hydration level in the first solvation shell suggesting that the hydronium ions drift away from the first solvation shell due to the shielding effect when solvation

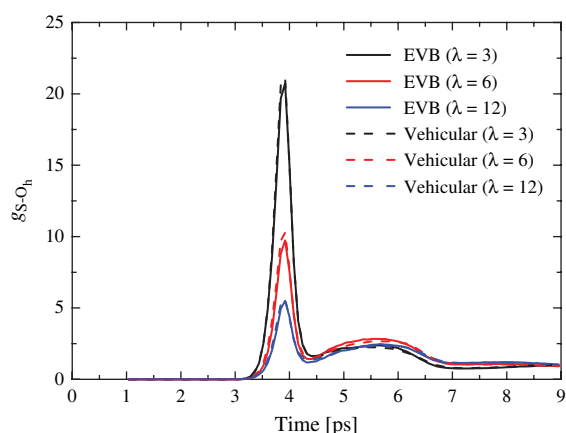


Figure 4. RDF of sulfur-hydronium oxygen for the EVB model (solid line) and vehicular model (dashed line) at hydration level 3, 6, and 12.

shells are saturated with water molecules. Interestingly, proton structure properties around the sulfonate groups are found to be very similar between the EVB model and vehicular model. It implies that the proton-hopping transport has a low impact on the proton dissociation from the sulfonate groups, i.e., protons are not enhanced to separate from the first solvation shells by the hopping mechanisms.

To understand the transport behavior of excess protons and water molecules, the mean square displacements (MSDs) of the hydronium ions and water molecules were calculated. Then, the diffusion coefficients, D , for the hydronium oxygen (O_h) and water oxygen (O_w) were determined from the MSDs using the Einstein-Smoluchowski equation written as

$$D = \lim_{t \rightarrow \infty} \frac{1}{6t} \langle |r(t) - r(0)|^2 \rangle \quad (9)$$

where t is the diffusion time, $r(t)$ is the position vector of particles.

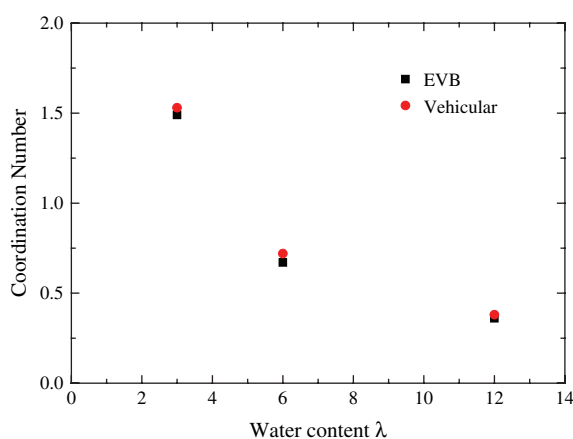


Figure 5. Coordination number of hydronium ions in the first solvation shell for the EVB model (black square) and vehicular model (red circle) at hydration level of 3, 6, and 12.

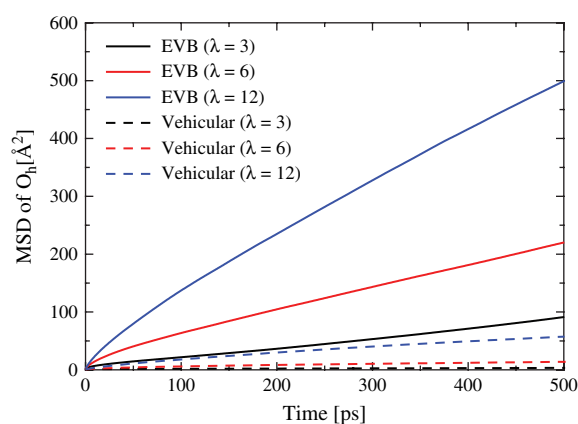


Figure 6. MSDs of hydronium ion oxygen for EVB model (solid line) and vehicular model (dashed line) at hydration level of 3, 6, and 12.

Figure 6 shows the MSDs of O_h for the EVB model (solid line) and vehicular model (dashed line) as a function of time for the water content of $\lambda = 3, 6$ and 12. The diffusion coefficients of hydronium ions were then estimated by the linear regime of these MSD curves from 300 to 500 ps at each water content. Figure 7 shows the diffusion coefficients of the hydronium ions for the EVB model and vehicular model from the present simulation. Experimental measurements²⁹ are also plotted in Figure 7. The diffusion coefficients of the vehicular model are in good agreement with the classical hydronium model¹⁷ previously reported by Devanathan et al.³⁰ for the studied hydration levels, testifying the validation of our vehicular model. With the EVB model, the simulation results were found to be in good agreement with experimental measurements²⁹ for the studied hydration levels, which significantly improve the validation of the simulation in comparison with previous studies.^{14–16, 31} In addition, the MSDs of the EVB model for the hydronium oxygen are larger and increase faster than those of the vehicular model as the water content increases as shown in Figure 6. Thus,

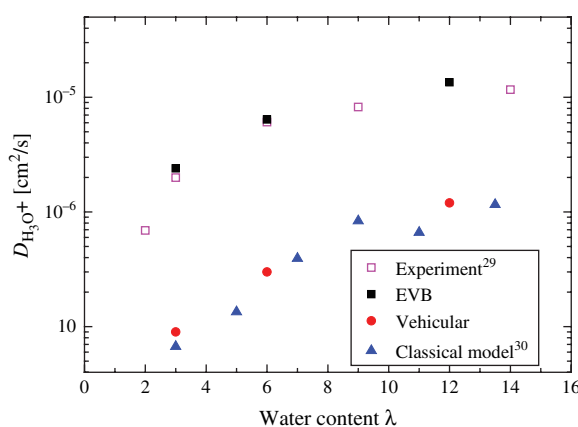


Figure 7. Diffusion coefficient of hydronium ions as a function of λ . Results are compared with experimental data taken from Ref. [29] and classical hydronium model taken from Ref. [30].

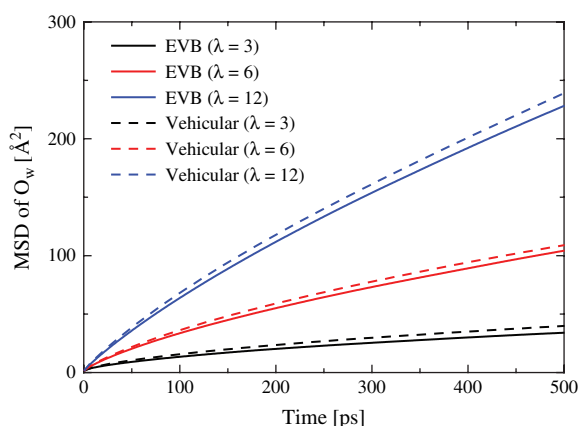


Figure 8. MSDs of water oxygen for EVB model (solid line) and vehicular model (dashed line) at hydration level of 3, 6, and 12.

the simulation results indicate that proton hopping mechanism becomes more significant at higher water contents. In fact, experimental results have also suggested that the proton-hopping mechanism becomes more important at higher water contents.^{32–34} Figure 8 shows the MSDs of O_w for the EVB model and vehicular model as a function of time for various water contents. Figure 9 shows the diffusion coefficients of water molecules for the EVB model and vehicular model from the present simulation calculated by the MSD slope and the experimental values of D_{H_2O} obtained from experimental measurements of nuclear magnetic resonance²⁹ (NMR) and quasi-elastic neutron scattering³⁵ (QENS). The simulation results are in good agreement with experimental data for the studied hydration levels. It should be noted that because the differences between D_{H_2O} values for the EVB model and vehicular model are small, a proton-hopping mechanism has a low impact on the diffusivity of water molecules for various water contents. Therefore, the results suggest that the vehicular mechanism dominates the electroosmotic transport of water molecules at the studied hydration levels.

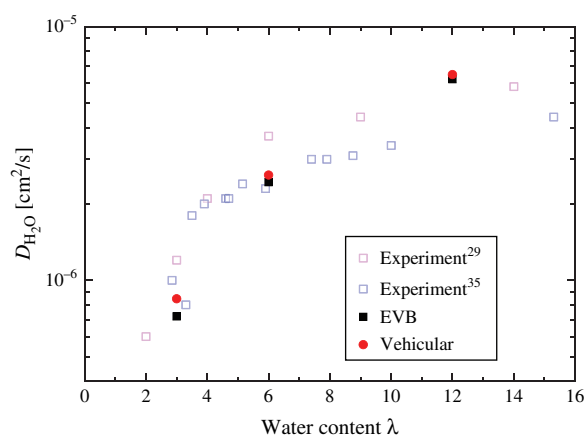


Figure 9. Diffusion coefficient of water molecules as a function of λ . Results are compared with experimental measurements of NMR taken from Ref. [29] and QENS taken from Ref. [35].

4. CONCLUSION

We have analyzed the structure and dynamics solvent molecules in hydrated Nafion at different hydration levels using a revised EVB model to investigate the proton solvation and transport properties. The revised EVB model successfully described proton transport through the Grotthuss hopping mechanism, giving significantly enhanced transport in comparison with the classical hydronium diffusion. It was found that sulfonate groups strongly interact with hydrated protons and water molecules and form the solvation shells. Our results show that the average number of protons in the first solvation shell (the coordination number n_{ch1}) decreases with increasing hydration level, namely that protons drift away from the first solvation shell due to the increase the shielding effect as the hydration level increases and the solvation shells are saturated with water molecules. Because the results of the structure analyses for the EVB model were found to be very similar to those for the vehicular model, it implies that the proton-hopping transport has a low impact on the proton dissociation from the sulfonate groups. In other words, proton separations from the sulfonate groups are not necessarily facilitated by the hopping mechanisms. The diffusion coefficients of protons were found to be consistent with the available experimental data, testifying the validation of the simulation models. It was found that a hopping-mechanism dominates the proton diffusion even at very low hydration and becomes more important factor contributing to the diffusion as the hydration level increases. The diffusion coefficients of water molecules were also found to be in good agreement with experimental data for the studied hydration levels and this result validates our calculation models. Diffusion analysis revealed that a proton-hopping mechanism has a small effect on the diffusivity of water molecules for the studied hydration levels because the differences between D_{H_2O} values for the EVB model and vehicular model are small. Thus, the results suggest that the vehicular mechanism dominates the electroosmotic transport of water molecules at the studied hydration levels.

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